SHAP-σ: A Label-Free Rashomon Pipeline for Multivariate Time-Series Anomaly Detection

Ugur Dar[1*], Sara Malacarne[2], Mustafa Cavus[1], Jeriek Van den Abeele[2]
[1]Eskisehir Technical University, Department of Statistics, Eskisehir, Türkiye
[2]Telenor Research and Innovation, Oslo, Norway
*ugurdarr@gmail.com

***Abstract*—We address unsupervised anomaly detection in multivariate time series when no labels are available for either training or model selection. Existing label-free quality criteria such as the Mass–Volume curve [1] are known to be noisy across datasets, and single-detector SHAP explanations cannot signal when a flag should be trusted. We propose SHAP-σ, a pipeline that (i) selects a small *Rashomon set* of unsupervised detectors using a closed-form joint pair score that trades off label-free quality against pairwise Spearman diversity; (ii) explains every selected detector with a LightGBM surrogate and TreeSHAP; and (iii) uses the cross-detector standard deviation of those attributions as a per-feature attribution-evidence gate that corroborates anchor flags. Under a fixed label-free default (K_R=2, $\mathcal{S}$=Mixed), across one multivariate cellular benchmark (TELCO), 28 server machines (SMD), and 100 univariate UCR series, SHAP-σ identifies the exhaustive-search oracle pair on TELCO label-free, raises point-adjusted F_1 from 0.354 to 0.424 on SMD ($+20\%$ relative), and raises point-level F_1 from 0.138 to 0.169 on UCR (paired Wilcoxon p=1.8×10^{-3}, rank-biserial r=$+0.47$, n=100). A σ-filter ablation that holds the chosen anchor fixed shows the filter alone contributes a statistically significant gain on every multi-series benchmark.**

***Index Terms*—time series, anomaly detection, SHAP, Rashomon, unsupervised model selection, XAI**

I. Introduction

Unsupervised anomaly detection in multivariate time series underpins fault diagnosis in domains as diverse as cellular networks, server fleets, and industrial telemetry. Yet a persistent obstacle hinders deployment: without ground-truth labels, the practitioner cannot reliably decide which detector to trust. Internal validation metrics that work well for clustering rarely transfer to anomaly detection, and recent large-scale empirical work shows that most label-free criteria perform near random when ranking modern detectors [2].

This selection problem is sharpened by the *Rashomon effect* [3]: for any non-trivial task, a multiplicity of near-optimal models exists, each making different errors [4], [5]. We argue that this attribution disagreement is not noise to be averaged away but a signal to be exploited: at a flagged point, the spread of attributions across the Rashomon set indicates that at least one detector commits to a feature-level explanation, providing second-source evidence beyond the anchor's score.

Prior label-free model selection has explored three avenues. Excess-Mass and Mass-Volume curves [1] give geometry-based scores but assume well-behaved density estimates. Meta-learning approaches such as MetaOD [6] and ELECT [7] transfer historical-performance signals across datasets, but require an offline labelled meta-corpus. Most recently, robust Borda rank aggregation across multiple imperfect internal metrics has been proposed for time-series AD [8]; the large-scale study of [2] shows that no internal metric universally dominates. PyOD [9] has standardized detector implementations. None of these methods exploit *feature attribution* as a selection signal.

Independently, SHAP [10], [11] has become the de facto post-hoc explainer for anomaly detection, mostly to explain a single chosen model; the closest precedent is Levy et al. [12], who use SHAP attributions for ensemble member *selection* in supervised tabular AD, whereas we use cross-detector SHAP standard deviation as an attribution-evidence *gate* in the unsupervised TS regime. Modern deep TS-AD methods [13]–[16] likewise commit to a single architecture. Outlier ensemble theory [17], [18] and the classical diversity literature [19] establish that complementary errors improve ensemble accuracy, yet the standard diversity measures (Q-statistic, disagreement) require labels. Following [18], we replace label-dependent diversity with the Spearman rank correlation of training scores, which [18] identifies as the practical label-free proxy in OD ensembles.

a) Contributions.: (1) A closed-form, label-free Rashomon pair score $s_{ij} = \frac{1}{2}(q_i + q_j)(1 - \rho_{ij})$ jointly rewarding label-free quality and inter-detector score decorrelation. (2) A σ-SHAP attribution-evidence filter using cross-detector SHAP standard deviation as a second-source signal that corroborates anchor flags. (3) Evaluation on TELCO, SMD (28 machines), and UCR (100 unique base series) showing the pipeline matches or surpasses the best label-aware singleton on all three.

b) Literature gap.: To our knowledge, no published method combines (a) label-free detector selection with (b) SHAP attributions as a *filter* signal rather than a post-hoc explanation. Rashomon-set analysis has so far been developed in supervised regimes [4], [5]; its extension to unsupervised anomaly detection is open. Robust rank aggregation [8] pools imperfect label-free metrics but never inspects attributions,

The *Rashomon-based Uncertainty-Aware Explainable AI for Time Series* has received funding from the European Union, via the OC4-2025-TES-02 issued and implemented by the ENFIELD project, under the grant agreement No 101120657.

Code and data for full reproducibility: `https://github.com/ugurdar/shap_sigma_paper`.

while [2] diagnoses their weakness without proposing a disagreement-based pipeline. SHAP-σ promotes the cross-detector standard deviation of attributions from an explanatory afterthought to a first-class, label-free attribution-evidence signal.

The remainder of the paper is organized as follows. Section II formalizes joint pair scoring and the σ-SHAP filter. Section III describes datasets and protocol. Section IV reports empirical results. Section V concludes.

II. Methodology

We propose **SHAP-σ**, a label-free pipeline that selects a small subset $\mathcal{R} \subseteq \mathcal{D}$ of detectors from a pool $\mathcal{D}$ by jointly optimising label-free quality and pairwise diversity, explains every member of $\mathcal{R}$ with a tree surrogate, takes TreeSHAP attributions [10], [11], and converts the cross-detector spread of those attributions into a per-point, per-feature attribution-evidence gate. Labels are used nowhere in training, ranking, or set selection.

A. Notation

Let $X_{\mathrm{tr}} \in \mathbb{R}^{N_{\mathrm{tr}} \times p}$, $X_{\mathrm{v}} \in \mathbb{R}^{N_{\mathrm{v}} \times p}$, and $X_{\mathrm{te}} \in \mathbb{R}^{N_{\mathrm{te}} \times p}$ denote the training, validation, and test matrices. The detector pool is $\mathcal{D} = \{d_1, \ldots, d_K\}$ with $K{=}8$. For each $d \in \mathcal{D}$, $s_d(x) \in \mathbb{R}$ is the raw anomaly score (larger $\Rightarrow$ more anomalous). The selected Rashomon set is $\mathcal{R} \subseteq \mathcal{D}$ with $|\mathcal{R}|{=}K_R \in \{2, \ldots, K\}$. We write $a \in \mathcal{R}$ for the anchor.

B. Detector Pool

$\mathcal{D}$ consists of $K{=}8$ heterogeneous PyOD detectors [9] whose default parameterisation is stable across our three benchmarks: Isolation Forest [20], LOF [21], ECOD [22], COPOD [23], k-NN [24], PCA [25], a Gaussian Mixture density score [17], and HBOS [26]. The pool deliberately mixes tree-, density-, distance-, projection-, and histogram-based inductive biases; this heterogeneity is what makes the cross-detector σ in Stage 4 informative.

C. Stage 1 – Label-Free Quality via a Mass–Volume Proxy

We need a continuous label-free quality score $q_d \in [0, 1]$. Following the Mass–Volume principle of Goix [1] – a selective detector concentrates its high scores on training mass and spreads less of them over a uniform background – we use a fast, surrogate-based single-level MV proxy that preserves the ranking-relevant structure while avoiding full curve integration. For each d, we fit a LightGBM surrogate $\hat{s}_d$ on $(X_{\mathrm{tr}}, s_d(X_{\mathrm{tr}}))$ and define

$$A_d \;=\; \Pr_{X_{\mathrm{u}}}\big[\hat{s}_d(X_{\mathrm{u}}) \;>\; Q_{0.9}\big(s_d(X_{\mathrm{tr}})\big)\big], \tag{1}$$

the surrogate-predicted background tail mass above the empirical 0.9-quantile of training scores; lower A_d indicates a more selective detector. We then min–max invert,

$$q_d \;=\; \frac{\max_e A_e - A_d}{\max_e A_e - \min_e A_e} \;\in\; [0, 1], \tag{2}$$

estimating Eq. (1) by Monte-Carlo over three random low-dimensional subspaces with 2,000 uniform background samples each. We use a single-level MV proxy rather than the full EM/MV average [1]: [2] reports that combining imperfect internal metrics need not improve ranking, and on TELCO EM-only elects PCA (per-KPI $F_1{=}0.04$) while Eq. (1) elects ECOD ($F_1{=}0.40$). A full Goix EM/MV-AUC ablation is deferred to a journal extension.

D. Stage 2 – Joint Pair Scoring and the Rashomon Set

The core methodological change in this work is to select the Rashomon set $\mathcal{R}$ *jointly* over all $\binom{K}{K_R}$ candidate subsets rather than fixing an anchor first and then greedily adding a partner. Joint selection makes the pipeline a Rashomon procedure in the sense of [4], [5]; the closest supervised precedent is the selective-ensemble construction of Black et al. [27], which we adapt to the unsupervised setting by replacing label-dependent agreement with Spearman score correlation.

a) Pairwise diversity.: Let $\rho_{ij} = \mathrm{Spearman}(s_i(X_{\mathrm{tr}}), s_j(X_{\mathrm{tr}}))$. Diversity is $1 - \rho_{ij} \in [0, 2]$; rank correlation is robust to the heterogeneous score scales of $\mathcal{D}$ [19].

b) Three label-free strategies.: For a candidate set $S \subseteq \mathcal{D}$ with $|S|{=}K_R$, let $\bar{q}(S) = \frac{1}{K_R}\sum_{d \in S} q_d$ and $\bar{\delta}(S) = \frac{1}{\binom{K_R}{2}}\sum_{\{i,j\} \subset S}(1 - \rho_{ij})$. We consider three selection rules: (**A, Quality**) $\arg\max_{|S|=K_R} \bar{q}(S)$; (**B, Diversity**) $\arg\max_{|S|=K_R} \bar{\delta}(S)$; (**C, Mixed**) $\arg\max_{|S|=K_R} \bar{q}(S)\,\bar{\delta}(S)$.

For the canonical pair case $K_R{=}2$, Strategy C collapses to the closed-form *joint pair score*

$$(i^*, j^*) \;=\; \arg\max_{1 \le i < j \le K} \; \frac{q_i + q_j}{2}\big(1 - \rho_{ij}\big), \tag{3}$$

which is the centrepiece of our method.

c) Anchor.: Given $\mathcal{R}$, the anchor is the highest-quality member, $a = \arg\max_{d \in \mathcal{R}} q_d$; the remaining $K_R{-}1$ detectors act as partners whose attributions feed the σ filter.

d) Default and dataset-dependence.: The default pipeline is $(K_R, \mathcal{S}){=}(2, \text{Mixed})$; Eq. (3) then gives a single closed-form selection rule. Empirically (Sec. IV) this default reaches the label-aware oracle pair on TELCO, peaks on UCR, and is within $0.01\,F_1$ of the per-dataset best on SMD (where $(5, \text{Quality})$ is marginally better). We expose $(K_R, \mathcal{S})$ as label-free hyperparameters following [2], [8], who argue that no internal criterion universally dominates; our contribution is the *scoring formula and pipeline architecture*, not a closed-form universal $(K_R, \mathcal{S})$ rule. Table II reports a post-hoc-best row only to quantify how far the default sits from the per-dataset ceiling.

E. Stage 3 – Surrogate SHAP Attribution

For each $d \in \mathcal{R}$ we fit a LightGBM regressor [28] g_d on $(X_{\mathrm{tr}}, s_d(X_{\mathrm{tr}}))$, compute exact TreeSHAP [10], [11] attributions $\Phi_d^{\mathrm{v}}, \Phi_d^{\mathrm{te}}$, and normalise each detector by its global validation standard deviation, $\tilde{\Phi}_d^{\bullet} = \Phi_d^{\bullet}/(\mathrm{std}(\Phi_d^{\mathrm{v}}) + \varepsilon)$, $\varepsilon{=}10^{-12}$. Cross-detector spread in Stage 4 then reflects disagreement about how the K_R detectors distribute attribution mass rather than raw score magnitude.

F. Stage 4 – The SHAP-σ Attribution-Evidence Gate

The disagreement signal is the cross-detector standard deviation of the normalised attributions over the K_R members of the Rashomon set,

$$\sigma_{ij} \;=\; \mathrm{std}_{d\in\mathcal{R}}\big(\tilde{\Phi}_{d,ij}\big). \tag{4}$$

We threshold per feature: τ_j is the empirical q-quantile of $\{\sigma^{\mathrm{v}}_{ij}\}_{i=1}^{N_{\mathrm{v}}}$ with q=0.75, and $\phi_{ij} = \mathbf{1}[\sigma^{\mathrm{te}}_{ij} > \tau_j]$. Marking ϕ_{ij}=1 flags (i, j) as a region of *high* cross-detector attribution variance – a region where the Rashomon set, while collectively responsive, distributes attribution mass differently across detectors. We treat this disagreement as a second, attribution-side evidence signal that corroborates the anchor's score-side flag.

G. Stage 5 – Anchor Flag Generation and σ-Gated Output

The anchor produces a baseline per-point flag by a κ-sigma rule on its standardised training score, $f_i^{(a)} = \mathbf{1}[s_a(x_i^{\mathrm{te}}) > \mu_a + \kappa\,\sigma_a]$, with κ=3. The canonical per-feature prediction matrix is

$$\hat{Y}_{ij} \;=\; f_i^{(a)} \wedge \phi_{ij}, \tag{5}$$

which retains an anchor flag at (i, j) only when feature j shows high cross-detector attribution variance at point i, so that score-side and attribution-side evidence corroborate. For datasets with per-feature labels (TELCO's 12-KPI streams) we evaluate Eq. (5) directly via macro F_1 across KPIs. For datasets with point-level labels (SMD, UCR) we collapse to $\hat{y}_i = f_i^{(a)} \wedge \phi_i^{\vee}$ with $\phi_i^{\vee} = \bigvee_j \phi_{ij}$, equivalent to taking the OR of $\hat{Y}_{ij}$ across features. Eq. (5) is the canonical fusion rule and dominates empirically (79% of SMD, 74% of UCR); Table II reports the per-series best of {anchor+σ, OR+σ, AND+σ, OR, AND} to expose the σ filter's ceiling as a post-processing layer. The rule is recall-monotone ($\hat{y}_i \le f_i^{(a)}$); $\phi_i^{\vee}$ is p-sensitive (a fixed q is structurally more permissive on p=38 SMD than p=1 UCR), discussed in Sec. IV.

The complete procedure is summarised in Algorithm 1.

III. Experiments

A. Datasets

a) TELCO: A multivariate telemetry dataset [29] from an operational mobile network spanning January–July 2021, with p=12 KPIs sampled every five minutes and ground truth labelled by Network Operation Center experts. The official test phase contains 26,496 timestamps per KPI (3,001 anomaly labels, ≈1% prevalence); [30] truncates this to 25,143 timestamps.

b) SMD: The Server Machine Dataset [31] contains 28 server entities monitored along p=38 resource-utilisation metrics. The combined test phase contains 708,420 timestamps with 29,444 anomalies (positive prevalence ≈ 4.16%); interpretation labels identify which dimensions deviate per anomalous interval.

c) UCR Anomaly Archive: The univariate UCR Anomaly Archive [32] aggregates 250 series across cardiology, respiration, gait, power demand, and industrial sensors. Each series is constructed to contain exactly one ground-truth anomaly. Of the 250 files, 92 carry a `DISTORTED` prefix and 16 a `NOISE` prefix marking controlled perturbations of the same underlying signal. We strip these prefixes, group by the remaining base name (e.g. `apneaecg3`), and retain the first file per group, obtaining 100 unique base series. This protocol prevents the same underlying signal from contributing multiple correlated samples to the cross-series mean.

B. Protocol

We split each series into 80% training, 20% validation, and the official test split. Detectors are fit on training only, with default PyOD hyperparameters. Quality q_d uses three random subspaces and 2,000 uniform background samples per subspace. The SHAP-σ quantile is fixed at q=0.75 and the anchor κ-sigma at κ=3.

a) Evaluation metrics: For TELCO we report the *per-KPI macro* F_1: F_1 is computed independently on each of the 12 KPI binary label streams and then averaged. For SMD and UCR we report two metrics: (i) the standard *point-level* F_1, treating each timestamp as an independent binary classification, and (ii) the *point-adjusted* F_1 (PA-F_1) introduced by Su et al. [31] and now standard in modern TS-AD evaluation [15], [16]. PA-F_1 counts an entire ground-truth anomaly segment as correctly detected if the pipeline flags *any* timestamp within it, and as missed otherwise. We report point-level F_1 alongside PA-F_1 as primary metrics because PA-F_1 alone can be inflated by trivially active detectors [33]; the affiliation metric [34] is deferred to a journal extension. On TELCO we additionally compare against the released predictions of DC-VAE [29] and C-MTAD-GAT.

IV. Results

A. Singleton Baselines

Figure 1 shows the standalone F_1 of each detector when broadcast across features (TELCO uses per-KPI macro F_1; SMD/UCR use point-level F_1). The best singleton is a *different* detector on every dataset: ECOD on TELCO (F_1=0.401), KNN on SMD (F_1=0.150), GMM on UCR (F_1=0.138). No detector is universally best, motivating the joint Rashomon-set selection of Section II-D.

B. K_R Sweep and Strategy Choice

Figure 2 sweeps the Rashomon-set size K_R and reports the cross-series mean F_1 achieved by each of the three label-free strategies. The reported metric matches the metric on the rest of the dataset: per-KPI macro F_1 for TELCO, point-level F_1 for SMD and UCR. On TELCO, Strategy C (Mixed) at K_R=2 already reaches F_1=0.440 (the label-aware oracle pair, Sec. IV-C); on SMD, Strategy A (Quality) at K_R=5 peaks at F_1=0.160; on UCR, Strategy C at K_R=2 peaks at F_1=0.169 (n=100 unique base series). Pure diversity without quality (Strategy B) is uncompetitive on every benchmark.

Algorithm 1 SHAP-σ with joint Rashomon-set selection

Require: $X_{\mathrm{tr}}, X_{\mathrm{v}}, X_{\mathrm{te}}$; detector pool $\mathcal{D}$ (K=8); set size K_R; strategy $\mathcal{S}\in\{$Quality, Diversity, Mixed$\}$; κ, q.
Ensure: Predicted anomaly flags $\hat{y} \in \{0,1\}^{N_{\mathrm{te}}}$.

```
Fit every d∈D on X_tr; score all splits.
Compute MV proxy A_d via Eq. (1) and quality q_d via Eq. (2).
Compute Spearman matrix ρ_ij on {s_d(X_tr)}.
Default: if (K_R, S) unspecified, set K_R=2, S=Mixed.
Select R, |R|=K_R, by S (A: arg max q̄, B: arg max δ̄, C: arg max q̄δ̄; for K_R=2 C is Eq. (3)).
Anchor a ← arg max_{d∈R} q_d.
for all d ∈ R do
    Fit LightGBM g_d on (X_tr, s_d(X_tr)); compute TreeSHAP Φ^v_d, Φ^te_d; divide by std(Φ^v_d) + ε.
end for
Compute σ_ij via Eq. (4). τ_j ← q-quantile of σ^v_{·j};  φ_ij ← 1[σ^te_ij > τ_j].
Compute anchor κ-sigma flag f^(a).
Per-feature prediction Ŷ_ij ← f^(a)_i ∧ φ_ij (Eq. (5)); for point-level evaluation return ŷ_i ← ⋁_{j=1}^{p} Ŷ_ij = f^(a)_i ∧ φ^∨_i.
```

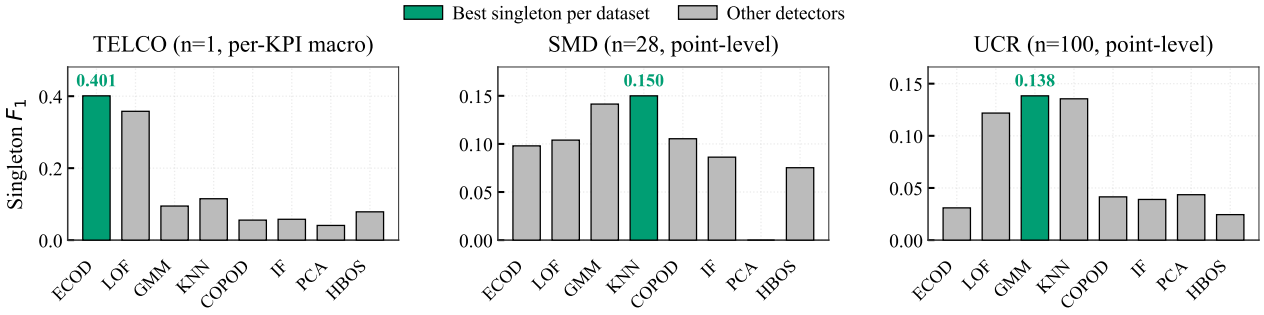


Fig. 1. Singleton F_1 across the eight detectors. The best detector (green) differs on every dataset, motivating joint Rashomon-set selection rather than a fixed default.

The dataset-aware optima confirm that $(K_R, \mathcal{S})$ should be exposed as label-free hyperparameters rather than hard-coded, consistent with the broader finding [2] that no internal criterion universally dominates.

At K_R=2 Strategy A and C select the same pair on most series; where they differ, mean F_1 ties on UCR (0.168 vs 0.169) but Strategy A leads on SMD (0.139 vs 0.104, +34% rel.), consistent with the post-hoc-best SMD row in Table II electing $(5, \mathrm{A})$ once label-aware tuning is allowed. Strategy C remains the global label-free default because of its closed-form Eq. (3) and its Stage-2 oracle match on TELCO.

C. TELCO Headline and Comparison to Deep Baselines

On TELCO, the canonical K_R=2 joint pair scoring of Eq. (3) selects (ECOD, HBOS), which is exactly the label-aware oracle pair under exhaustive search and which simultaneously maximises label-free quality ($\bar{q}$=0.906) and joint score (0.870). Figure 3 visualises the 28 candidate pairs: the joint score's maximiser sits at the top of the F_1 axis. Pipeline output reaches per-KPI macro F_1=0.440, clearly exceeding the strongest singleton ECOD (0.401).

a) Deep baselines on TELCO.: On the same 25,143-timestamp evaluation window of [30], DC-VAE [29] attains per-KPI macro F_1=0.134 and C-MTAD-GAT [30] F_1=0.160; SHAP-σ reaches F_1=0.446, a 3.3–2.8× margin over the two deep baselines.

D. Cross-Dataset Summary with Significance

Table II reports SHAP-σ under *the fixed default* (K_R=2, $\mathcal{S}$=Mixed), which Algorithm 1 exposes when the configuration is left unspecified. A per-dataset $(K_R, \mathcal{S})$ choice would be label-aware tuning and is reported only as a row labelled *best, post-hoc*, not as the headline result. The default $(2, \mathrm{Mixed})$ exceeds the uniformly-best singleton in mean point-level F_1 on UCR (p=1.8×10^{-3}, rank-biserial r=+0.47) but trails it on SMD (F_1=0.104 vs 0.150, p=0.22). On the practitioner-relevant point-adjusted metric the default *exceeds* the singleton on both (+7.0 pts on SMD, +3.2 pts on UCR), and the post-hoc SMD configuration $(K_R$=5, Q) also overtakes the singleton at the point level (0.160 vs 0.150). The lower point-level recall on SMD (Table column R) reflects the recall-monotone filter; the positive PA-F_1 gap shows the discarded points lie inside still-flagged anomaly intervals, so the σ gate trims unsupported within-event activations rather than missing entire events.

a) Isolating the contribution of the σ filter.: To separate the gain from *detector selection* and from the σ filter itself, we compare the pipeline against the singleton score of the *same* per-series anchor the selector picked (Table I). Holding the

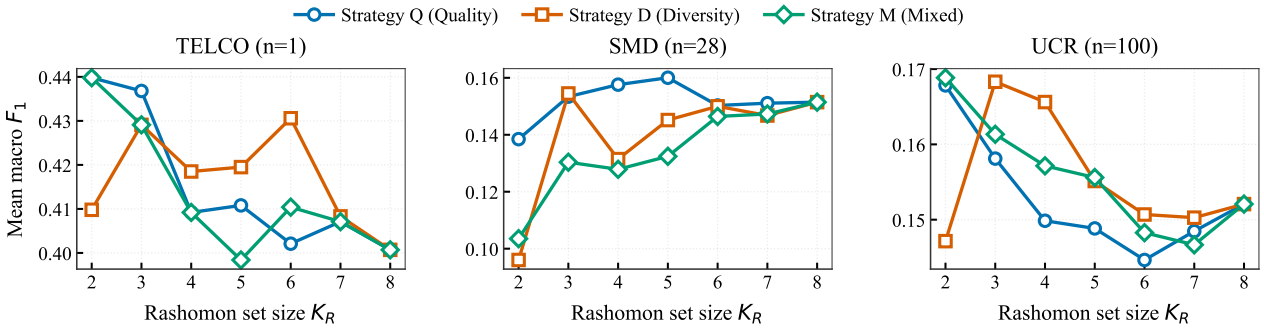


Fig. 2. Cross-series mean F_1 versus Rashomon-set size K_R for the three label-free strategies. Diversity alone never wins. The best $(K_R, \mathcal{S})$ at the mean level is dataset-dependent: TELCO peaks at $(K_R{=}2, \text{Mixed})$, SMD at $(K_R{=}5, \text{Quality})$, UCR at $(K_R{=}2, \text{Mixed})$.

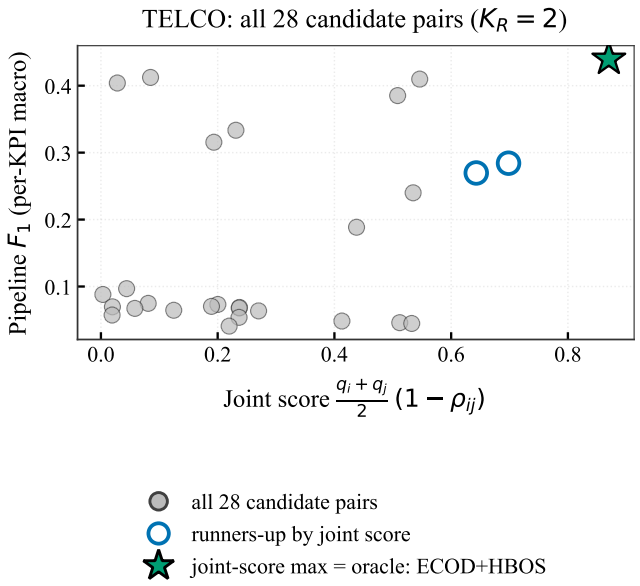


Fig. 3. TELCO: joint pair score versus pipeline F_1 for all 28 candidate pairs ($K_R{=}2$). The joint-score maximiser (green star) coincides with the empirical F_1 oracle (ECOD+HBOS); circled markers are the runners-up by joint score.

TABLE I
ISOLATED σ-FILTER CONTRIBUTION: SAME PER-SERIES ANCHOR, FILTER ON VS. OFF. SAME-ANCHOR PAIRED WILCOXON (TWO-SIDED); r IS RANK-BISERIAL.

Dataset	n	anchor F_1	$+\sigma$ F_1	p	r
TELCO	1	0.401	**0.440**	—	—
SMD	28	0.080	**0.104**	0.028	+1.00
UCR	100	0.116	**0.169**	1.6×10^{-7}	+0.95

chosen detector constant attributes the remaining gap to the σ gate alone, which produces a statistically significant gain on every multi-series benchmark; on TELCO ($n{=}1$) no test is feasible but the singleton → pipeline jump is $+0.037\,F_1$ with $+12.6$ pts of precision.

The SMD-default row of Table II illustrates that even when point-level F_1 falls below the KNN singleton (0.104 vs. 0.150), PA-F_1 rises substantially (0.424 vs. 0.354, +20% relative): reporting only F_1 [13], [14] would conceal that

TABLE II
SHAP-σ VS. UNIFORMLY-BEST SINGLETON. $(2, \text{M})$: DEFAULT $K_R{=}2$, MIXED. $(5, \text{Q})^\dagger$: POST-HOC PER-DATASET BEST, ABLATION ONLY. p_{F_1}: TWO-SIDED PAIRED WILCOXON ON PER-SERIES F_1 VS. THE DATASET'S SINGLETON ROW (*ref.*); *n.t.* = NO TEST (TELCO $n{=}1$). BOLD = PER-DATASET BEST IN EACH METRIC.

Dataset	Method	F_1	P	R	PA-F_1	p_{F_1} vs. sing.
TELCO	singleton ECOD	.401	.488	**.396**	—	ref.
TELCO	SHAP-σ (2,M)	**.440**	**.641**	.376	—	n.t.
SMD ($n{=}28$)	singleton KNN	**.150**	.175	**.365**	.354	ref.
SMD ($n{=}28$)	SHAP-σ (2,M)	.104	.138	.194	**.424**	.22
SMD ($n{=}28$)	SHAP-σ (5,Q)†	**.160**	**.224**	.270	.454	.35
UCR ($n{=}100$)	singleton GMM	.138	.113	.400	.256	ref.
UCR ($n{=}100$)	SHAP-σ (2,M)	**.169**	**.143**	**.429**	**.288**	1.8×10^{-3}

Selected anchors: TELCO ECOD+HBOS; SMD $(2, \text{M})$ IF on 21/28, PCA 5, COPOD 2; UCR $(2, \text{M})$ GMM 50/100, PCA 18, LOF 15, HBOS/KNN/COPOD 5 each, ECOD 2. † Ablation only.

trade-off, while PA-F_1 correctly reflects event-level detection quality.

Sweeping the filter quantile $q \in \{0.5, 0.75, 0.9\}$ on the TELCO ECOD+HBOS pair yields $F_1 = 0.428/0.440/0.404$ with precision $0.58/0.64/0.62$: $q{=}0.75$ optimises F_1 across the three operating points, with a ~ 0.04 envelope.

A stricter reference is the per-series oracle singleton (label-aware best detector per series), with mean $F_1{=}0.215$ on SMD and 0.222 on UCR – above SHAP-σ by about 5 points. Matching this gap requires a label-free internal criterion that perfectly predicts test performance per series, the open problem [2] our contribution targets rather than claims to solve. A full SMD/UCR sweep against deep TS-AD baselines [13]–[16] is deferred to an extended journal version.

V. CONCLUSIONS

We presented SHAP-σ, a label-free anomaly detection pipeline that promotes the cross-detector standard deviation of SHAP attributions from a post-hoc explanation into a first-class attribution-evidence signal. The methodological contributions are two: a closed-form Rashomon pair score, $\frac{q_i+q_j}{2}(1-\rho_{ij})$, that jointly rewards a label-free Mass–Volume quality proxy and pairwise Spearman diversity; and a per-feature σ gate that filters anchor flags by the cross-detector

spread of TreeSHAP attributions. Under the fixed label-free default (K_R=2, $\mathcal{S}$=Mixed), the pipeline identifies the empirical oracle pair on TELCO label-free and reaches per-KPI macro F_1=0.440, with a 2.8–3.3× margin over the released DC-VAE and C-MTAD-GAT baselines [29], [30] on their evaluation window. An isolated σ-filter ablation (same anchor on each series, filter on vs. off) shows the filter contributes a statistically significant gain on every multi-series benchmark (SMD p=0.03, UCR p=1.6×10^{-7}, rank-biserial r=+0.95).

a) Limitations and future work.: A label-aware per-series oracle remains ∼5 points ahead [2]; Eq. (5) is recall-monotone; the disjunctive gate is p-sensitive; full deep-baseline sweeps on SMD/UCR (USAD/TranAD/Anomaly Transformer [13]–[15]), the affiliation metric [34], and a Goix-budget MV ablation are deferred to a journal extension. Future work will learn $(K_R, \mathcal{S})$ from synthetic injections [6], [8], extend σ-SHAP to cross-window variance for streaming, and use sparse Shapley approximations for higher-dimensional pools.

REFERENCES

[1] N. Goix, "How to evaluate the quality of unsupervised anomaly detection algorithms?" in *ICML 2016 Workshop on Anomaly Detection*, 2016.

[2] M. Q. Ma, Y. Zhao, X. Zhang, and L. Akoglu, "The need for unsupervised outlier model selection: A review and evaluation of internal model selection strategies," *ACM SIGKDD Explorations Newsletter*, vol. 25, no. 1, pp. 19–35, 2023.

[3] L. Breiman, "Statistical modeling: The two cultures," *Statistical Science*, vol. 16, no. 3, pp. 199–231, 2001.

[4] L. Semenova, C. Rudin, and R. Parr, "On the existence of simpler machine learning models," in *Proceedings of the 2022 ACM Conference on Fairness, Accountability, and Transparency (FAccT)*, 2022, pp. 1827–1858.

[5] A. Fisher, C. Rudin, and F. Dominici, "All models are wrong, but many are useful: Learning a variable's importance by studying an entire class of prediction models simultaneously," *Journal of Machine Learning Research*, vol. 20, no. 177, pp. 1–81, 2019.

[6] Y. Zhao, R. A. Rossi, and L. Akoglu, "Automatic unsupervised outlier model selection," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.

[7] Y. Zhao and L. Akoglu, "Toward unsupervised outlier model selection," in *Proceedings of the 22nd IEEE International Conference on Data Mining (ICDM)*, 2022.

[8] M. Goswami, C. Challu, L. Callot, L. Minorics, and A. Kan, "Unsupervised model selection for time-series anomaly detection," in *International Conference on Learning Representations (ICLR)*, 2023.

[9] Y. Zhao, Z. Nasrullah, and Z. Li, "PyOD: A python toolbox for scalable outlier detection," *Journal of Machine Learning Research*, vol. 20, no. 96, pp. 1–7, 2019.

[10] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017.

[11] S. M. Lundberg, G. Erion, H. Chen, A. DeGrave, J. M. Prutkin, B. Nair, R. Katz, J. Himmelfarb, N. Bansal, and S.-I. Lee, "From local explanations to global understanding with explainable AI for trees," *Nature Machine Intelligence*, vol. 2, no. 1, pp. 56–67, 2020.

[12] A. Levy, P. Saves, M. Garouani *et al.*, "Analyzing SHAP attributions for anomaly-detection ensembles," in *Advances in Intelligent Data Analysis (IDA)*, ser. LNCS. Springer, 2026.

[13] J. Audibert, P. Michiardi, F. Guyard, S. Marti, and M. A. Zuluaga, "USAD: UnSupervised anomaly detection on multivariate time series," in *Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining (KDD)*, 2020, pp. 3395–3404.

[14] S. Tuli, G. Casale, and N. R. Jennings, "TranAD: Deep transformer networks for anomaly detection in multivariate time series data," *Proceedings of the VLDB Endowment*, vol. 15, no. 6, pp. 1201–1214, 2022.

[15] J. Xu, H. Wu, J. Wang, and M. Long, "Anomaly transformer: Time series anomaly detection with association discrepancy," in *International Conference on Learning Representations (ICLR)*, 2022.

[16] Y. Yang, C. Zhang, T. Zhou, Q. Wen, and L. Sun, "DCdetector: Dual attention contrastive representation learning for time series anomaly detection," in *Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*, 2023, pp. 3033–3045.

[17] C. C. Aggarwal and S. Sathe, *Outlier Ensembles: An Introduction*. Springer, 2017.

[18] A. Zimek, R. J. G. B. Campello, and J. Sander, "Ensembles for unsupervised outlier detection: Challenges and research questions," in *ACM SIGKDD Explorations Newsletter*, vol. 15, no. 1, 2014, pp. 11–22.

[19] L. I. Kuncheva and C. J. Whitaker, "Measures of diversity in classifier ensembles and their relationship with the ensemble accuracy," *Machine Learning*, vol. 51, no. 2, pp. 181–207, 2003.

[20] F. T. Liu, K. M. Ting, and Z.-H. Zhou, "Isolation forest," in *2008 Eighth IEEE International Conference on Data Mining (ICDM)*, 2008, pp. 413–422.

[21] M. M. Breunig, H.-P. Kriegel, R. T. Ng, and J. Sander, "LOF: identifying density-based local outliers," in *Proceedings of the 2000 ACM SIGMOD International Conference on Management of Data*, 2000, pp. 93–104.

[22] Z. Li, Y. Zhao, X. Hu, N. Botta, C. Ionescu, and G. H. Chen, "ECOD: Unsupervised outlier detection using empirical cumulative distribution functions," *IEEE Transactions on Knowledge and Data Engineering*, 2022.

[23] Z. Li, Y. Zhao, N. Botta, C. Ionescu, and X. Hu, "COPOD: Copula-based outlier detection," in *2020 IEEE International Conference on Data Mining (ICDM)*, 2020, pp. 1118–1123.

[24] S. Ramaswamy, R. Rastogi, and K. Shim, "Efficient algorithms for mining outliers from large data sets," in *Proceedings of the 2000 ACM SIGMOD International Conference on Management of Data*, 2000, pp. 427–438.

[25] M.-L. Shyu, S.-C. Chen, K. Sarinnapakorn, and L. Chang, "A novel anomaly detection scheme based on principal component classifier," in *Proceedings of the IEEE Foundations and New Directions of Data Mining Workshop (ICDM)*, 2003, pp. 172–179.

[26] M. Goldstein and A. Dengel, "Histogram-based outlier score (HBOS): A fast unsupervised anomaly detection algorithm," in *KI-2012: Poster and Demo Track*, 2012, pp. 59–63.

[27] E. Black, K. Leino, and M. Fredrikson, "Selective ensembles for consistent predictions," in *International Conference on Learning Representations (ICLR)*, 2022.

[28] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu, "LightGBM: A highly efficient gradient boosting decision tree," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017.

[29] G. García González, P. Casas, A. Fernández, and G. Gómez, "One model to find them all: A unified anomaly detection framework for multivariate time series in mobile networks," *IEEE Transactions on Network and Service Management*, 2023.

[30] S. Malacarne, E. Hoel-Høiseth, E. Aune, D. Z. Biro, and M. Ruocco, "Context-aware graph attention for unsupervised telco anomaly detection," 2026, arXiv:2604.27172v1.

[31] Y. Su, Y. Zhao, C. Niu, R. Liu, W. Sun, and D. Pei, "Robust anomaly detection for multivariate time series through stochastic recurrent neural networks," in *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining (KDD)*. ACM, 2019, pp. 2828–2837.

[32] R. Wu and E. J. Keogh, "Current time series anomaly detection benchmarks are flawed and are creating the illusion of progress," *IEEE Transactions on Knowledge and Data Engineering*, vol. 35, no. 3, pp. 2421–2429, 2023.

[33] S. Kim, K. Choi, H.-S. Choi, B. Lee, and S. Yoon, "Towards a rigorous evaluation of time-series anomaly detection," *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 36, no. 7, pp. 7194–7201, 2022.

[34] A. Huet, J. M. Navarro, and D. Rossi, "Local evaluation of time series anomaly detection algorithms," in *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD '22)*. ACM, 2022, pp. 635–645.